3D-LLDM: Label-Guided 3D Latent Diffusion Model for Improving High-Resolution Synthetic MR Imaging in Hepatic Structure Segmentation

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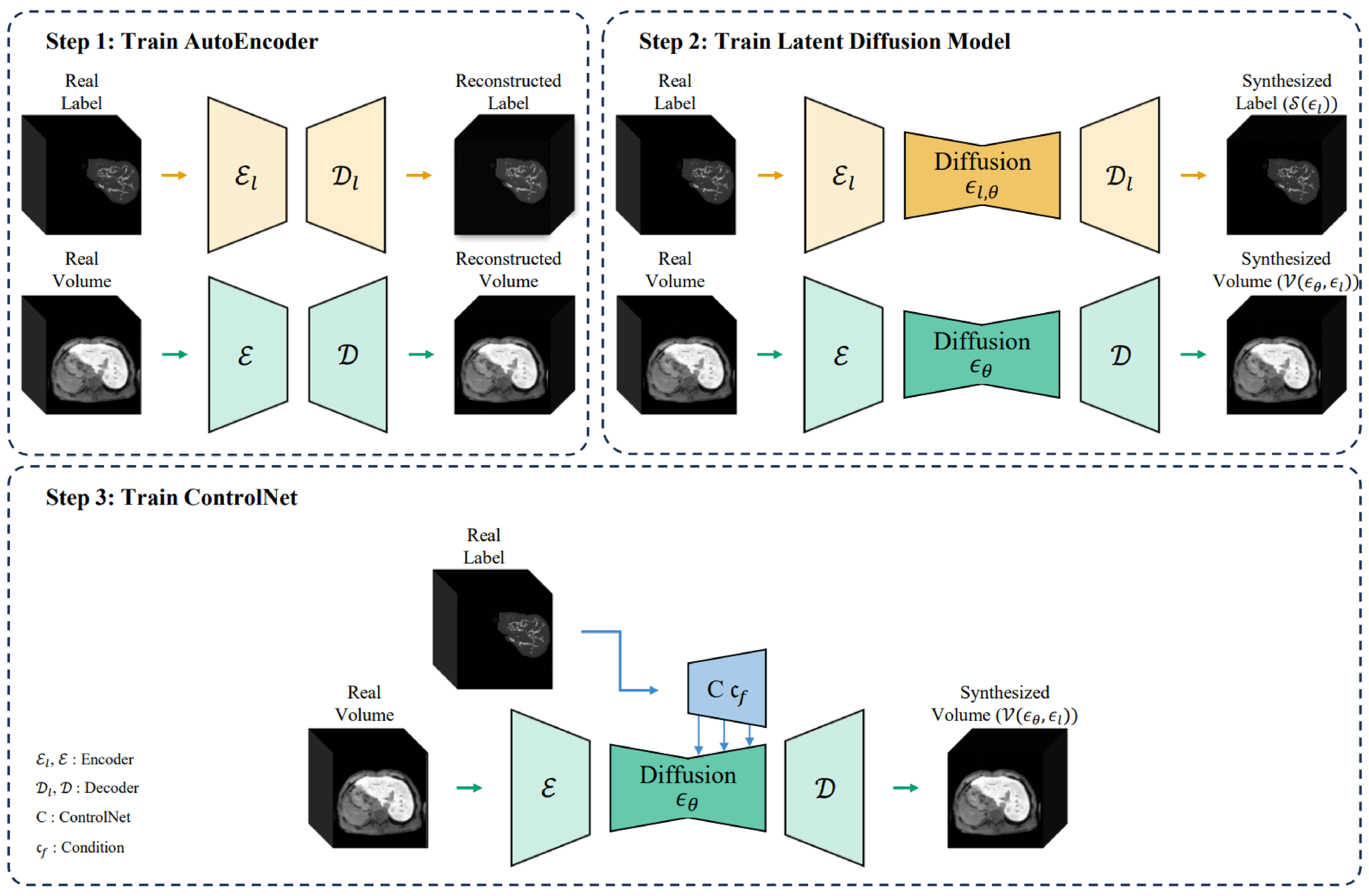
Introduction

- ✓ **Challenge:** Data scarcity and lack of anatomical integrity in synthetic 3D medical volumes.
- ✓ **Proposed: 3D-LLDM**, a label-guided diffusion framework for high-resolution MR synthesis using **ControlNet** [1].
- ✓ **Achievement:** Produced spatially consistent volumes and masks from 720 clinical scans (SMC).
- ✓ **Performance:** Achieved **FID 28.31** and improved HCC segmentation Dice score by **11.15%**.

Methods



3D-LLDM is a 2-stage framework designed to generate high-resolution synthetic medical volumes paired with spatially accurate labels.

Architecture

- **Autoencoders** $(\mathcal{E}, \mathcal{D})$**[2]:** Two separate pairs learn to compress/reconstruct labels and volumes.
- **Latent Diffusion (LDM)**[3]:
  - **Label LDM:** Learns the distribution of anatomical structures.
  - **ControlNet**[1]**:** Generates medical volumes conditioned on label geometry.

Training

- Stage 1: Train Autoencoders for both labels and volumes.
- Stage 2: Train the Label LDM and ControlNet using real label latents(z_l) for maximum alignment and stability.

$$\mathcal{L}_c = \mathbb{E}[\|\epsilon - \epsilon_\theta(z_t, t, z_l)\|_2^2]$$

Inference (Data Synthesis)

Generates perfectly aligned {Label, Volume} pairs from random noise:

1. Noise → Label LDM → Synthetic Label ($\mathcal{S}$)
2. $\mathcal{S}$ → ControlNet → Synthetic Volume ($\mathcal{V}$)

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References

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[2] L. P. Cinelli, M. A. Marins, E. A. B. Silva, and S. L. Netto, "Variational autoencoder," in Variational methods for machine learning with applications to deep networks, 2021, pp. 111-149.
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Results

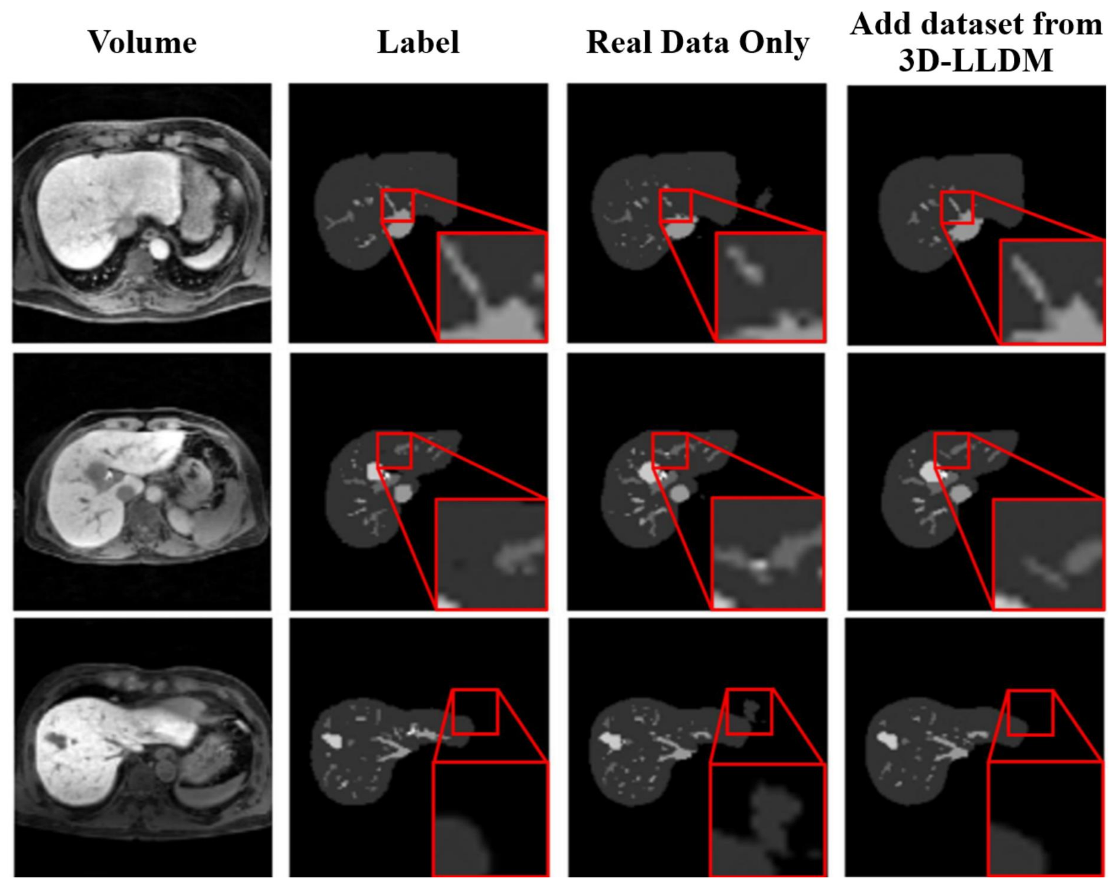
CNN Model	Segmentation	R	R + S	Improvement
U-Net	Liver-Only	0.9650	**0.9662**	+0.124%
	Vein-Only	0.7905	**0.8667**	+9.640%
	HCC-Only	0.7334	**0.8152**	**+11.153%**
	Multi-Class	0.6968	**0.7014**	+0.660%
ResUNet	Liver-Only	0.9633	**0.9634**	+0.010%
	Vein-Only	0.7197	**0.7902**	+9.796%
	HCC-Only	0.7108	**0.7646**	+7.569%
	Multi-Class	0.6601	**0.6652**	+0.773%
WideResUNet	Liver-Only	**0.9657**	**0.9661**	+0.041%
	Vein-Only	0.7230	**0.7989**	**+10.498%**
	HCC-Only	0.7251	**0.7857**	+8.357%
	Multi-Class	0.6961	**0.7020**	+0.848%
DynUNet	Liver-Only	0.9541	**0.9736**	+2.044%
	Vein-Only	0.6958	**0.7540**	+8.365%
	HCC-Only	0.7002	**0.7674**	+9.597%
	Multi-Class	0.6983	**0.7340**	+5.112%
VNet	Liver-Only	0.9289	**0.9677**	+4.177%
	Vein-Only	0.6908	**0.7212**	+4.401%
	HCC-Only	0.6350	**0.6825**	+7.480%
	Multi-Class	0.5584	**0.5937**	+6.322%
Overall Mean Dice	Liver-Only	0.9544	**0.9674**	+1.362%
	Vein-Only	0.7240	**0.7862**	+8.591%
	HCC-Only	0.7009	**0.7631**	+8.874%
	Multi-Class	0.6619	**0.6793**	+2.629%

Quantitative Performance

Synthetic Data Augmentation (Real + Synthetic) significantly outperforms Real-only training.

- Average performance gain across all tasks and architectures.
- HCC-Only: Highest individual gain (+11.15% for U-Net) and overall improvement (+8.87% avg).
- Vein-Only: Average improvement of +8.59%.

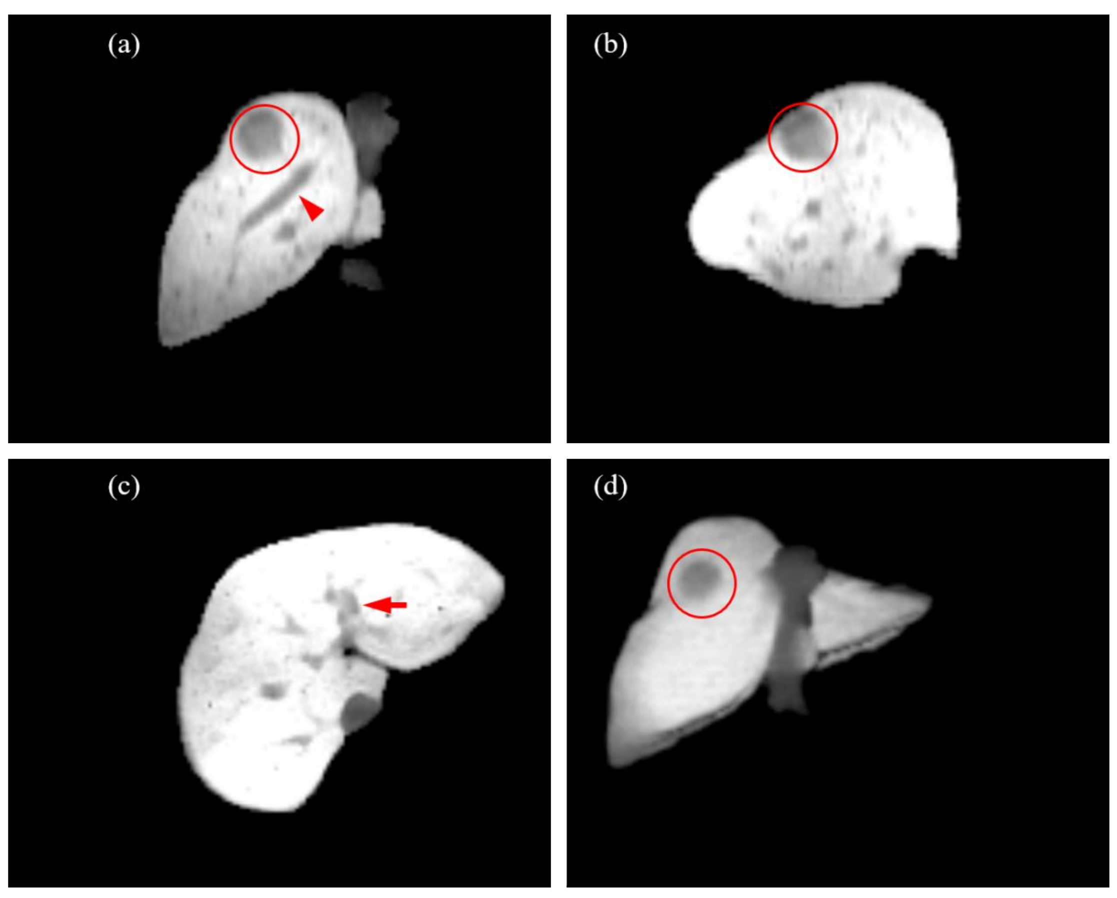
** Best and 2nd-best results are highlighted.*



Qualitative Comparison

Visual enhancement in anatomical continuity with R+S training.

- Models trained with Real + 3D-LLDM data show superior continuity in vessel segmentation.
- Discontinuous appearance of the middle hepatic vein in "Real-only" results is successfully recovered in "Real+Synthetic" results.
- Segmentation from R+S training align more closely with Ground Truth.



High-Quality 3D Synthesis

3D-LLDM generates anatomically consistent, high-resolution MR volumes.

- High-fidelity details preserved across Coronal, Sagittal, and Axial planes.
- Tumors: Precisely modeled within the volumetric data.
- Vascular Structures: Clear representation of right hepatic and left portal veins.
- Utility: Synthetic volumes provide realistic clinical features for robust model training.

Conclusion

- **Proposed 3D-LLDM:** A novel framework for generating high-resolution synthetic MR volumes with spatially aligned segmentation labels.
- **Spatial Guidance:** By leveraging **ControlNet**, our model ensures anatomical consistency, significantly outperforming conventional generative models with the **lowest FID score**.
- **Performance Gain:** Demonstrated superior segmentation accuracy for both liver and multi-class tasks, effectively mitigating **data scarcity** in medical imaging.
- **Future Work:** We aim to extend 3D-LLDM to diverse modalities and explore **domain adaptation** for enhanced generalization.